Multiscale Salient Alignment Learning for Remote-Sensing Image–Text Retrieval

Yaxiong Chen, Jinghao Huang, Xiaoyu Li, Shengwu Xiong, and Xiaoqiang Lu, *Senior Member, IEEE*

***Abstract*— *Remote-sensing image–text* (RSIT) retrieval involves the use of either textual descriptions or remote-sensing images (RSI) as queries to retrieve relevant RSIs or corresponding text descriptions. Many traditional cross-modal RSIT retrieval methods tend to overlook the importance of capturing salient information and establishing the prior similarity between RSIs and texts, leading to a decline in cross-modal retrieval performance. In this article, we address these challenges by introducing a novel approach known as multiscale salient image-guided text alignment (MSITA). This approach is designed to learn salient information by aligning text with images for effective cross-modal RSIT retrieval. The MSITA approach first incorporates a multiscale fusion module and a salient learning module to facilitate the extraction of salient information. In addition, it introduces an image-guided text alignment (IGTA) mechanism that uses image information to guide the alignment of texts, enabling the effective capture of fine-grained correspondences between RSI regions and textual descriptions. In addition to these components, a novel loss function is devised to enhance the similarity across different modalities and reinforce the prior similarity between RSIs and texts. Extensive experiments conducted on four widely adopted RSIT datasets affirm that the MSITA approach significantly enhances cross-modal RSIT retrieval performance in comparison to other state-of-the-art methods.**

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***Index Terms*— Cross-modal retrieval, image-guided text alignment (IGTA), prior similarity, salient learning.**

I. Introduction

In RECENT years, the proliferation of multimedia devices has led to an explosive growth in the volume of videos, images, and text data [1], [2], [3], [4]. This surge in data has generated significant interest in bridging the gap between vision and language, prompting extensive exploration of the relationship between the two [5], [6], [7], [8], [9]. Simultaneously, advancements in remote-sensing technology have driven the rapid expansion of multimodal remote-sensing data [10]. Remote-sensing imagery often exhibits considerable variation in object sizes within the same scene due to differences in acquisition altitudes [11], [12]. Furthermore, RSI can encompass a broad range of backgrounds and contexts that may be unrelated to the intended focus. This background content can impede the effective extraction of relevant information from RSI, making the retrieval of valuable data within large-scale RSI datasets a crucial challenge [13], [14], [15]. RSI retrieval can be categorized into two main types: single-modal RSI retrieval and cross-modal RSI retrieval. The former involves retrieving similar RSIs within the same modality [16], [17]. For instance, Demir and Bruzzone [18] introduced nonlinear hashing techniques to enhance retrieval efficiency and evaluated them in terms of accuracy and speed. Chen et al. [19] proposed an innovative retrieval method using hash codes to further improve retrieval efficiency. Conversely, cross-modal RSI retrieval encompasses RSI searches using queries and databases from different modalities [20], [21]. The interaction between different modalities in the remote-sensing domain has garnered significant attention from [15], [22], and [23]. For example, Zhen et al. [24] devised a similarity computation approach that leveraged pretrained models for images and audio, projecting features from different modalities into a shared space and evaluating retrieval correctness using labels. Faghri et al. [25] introduced a self-supervised approach aimed at enhancing the joint embedding of visual and textual semantics by incorporating challenging samples. Overall, the intersection of remote-sensing imagery, multimodal data, and cross-modal retrieval presents a rich landscape for research and innovation in effectively harnessing and understanding large-scale RSI datasets.

Cross-modal retrieval presents a significant challenge in learning joint embeddings and accurately assessing the similarity between images and texts due to the inherent disparities across modalities. Recent advancements in deep-learning retrieval methods have allowed for direct computation of

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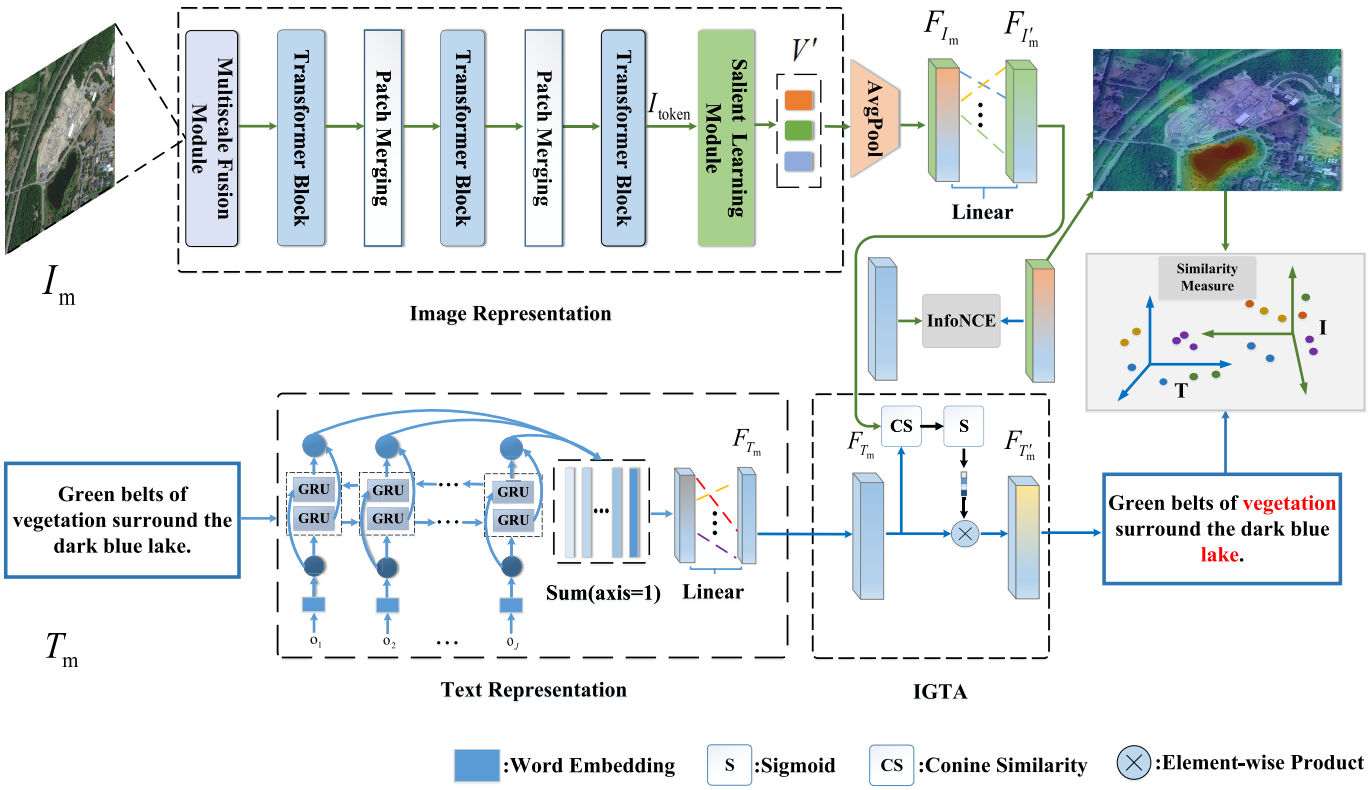


Fig. 1. Overview of the MSITA framework. The input of the MSITA framework is RSIT pairs. The framework consists of an RSI branch and a text branch. The RSI branch consists of a multiscale fusion module, three transformer blocks, two patch mergings, a sequence salient selection block, and a fully connected layer. The text branch consists of a bidirectional GRU, MLP, and IGTA. The overall objective function contains a bidirectional triplet term and an InfoNCE term.

similarity between images and texts [26], [27], [28]. However, when applying these approaches to cross-modal RSIT retrieval [29], [30], [31], several persistent challenges emerge. First, despite the remarkable progress achieved through multiscale structures in extracting information from remote-sensing images (RSIs) [32], [33], [34], there remains a difficulty in capturing subtle differences among small targets and effectively addressing regions lacking discriminative features. Second, traditional cross-modal remote-sensing methods have typically relied on rough alignment of texts with RSIs, resulting in the loss of essential text information associated with the image. The challenge here lies in leveraging image information to guide more effective text alignment. Last, the existing loss functions in cross-modal remote-sensing methods mainly gauge the similarity between the retrieval feature vectors of texts and RSIs. This approach overlooks the prior similarity between texts and RSIs. Consequently, finding a way to exploit this prior similarity to enhance the similarity between the retrieval feature vectors of texts and RSIs represents another challenge.

In response to these challenges, this article introduces a novel approach called the *multiscale salient image-guided text alignment* (MSITA) method. MSITA addresses cross-modal RSIT retrieval by focusing on capturing salient information, optimizing image-guided text alignment (IGTA), and learning the prior similarity between RSIs and texts. An overview of the MSITA framework is shown in Fig. 1. The MSITA framework comprises two main branches: the RSI branch and the text branch. The RSI branch includes modules for multiscale fusion, transformer blocks, patch merging, sequence salient selection, and fully connected layers. The text branch incorporates a bidirectional GRU, multilayer perceptron (MLP), and IGTA. To capture subtle differences among small targets, the multiscale fusion module and salient region learning module are devised to facilitate multiscale information learning and salient information extraction. To guide text alignment effectively with image information, the IGTA component is designed to explore fine-grained correspondences between RSI regions and texts. To preserve the prior similarity between texts and RSIs, a novel objective function is introduced, which includes bidirectional triplet and InfoNCE terms. This function expands the distance between instances and their corresponding negative instances while reducing the distance between instances and their positive instances. This is achieved by considering the similarity between $F_{I'_m}$ and $F_{T'_m}$ and the similarity between F_{I_m} and F_{T_m}. Extensive experiments conducted on four datasets demonstrate that the MSITA framework consistently outperforms other compared approaches in the context of RSIT retrieval. This research seeks to address critical challenges in cross-modal RSIT retrieval and introduces a promising framework for advancing the state of the art in this field.

The contributions of MSITA are outlined in the following aspects.

1) A novel cross-modal RSIT retrieval framework is designed to learn salient information and fine-grained

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correspondence relationships to tackle the issue of RSIs and text alignment. Moreover, the IGTA module is designed to provide image information to guide text alignment.

2) To better perverse the similarity of different modalis and reinforce the prior similarity between RSIs and texts, a novel objective function is designed to compute the similarity for retrieval feature vectors and perform the similarity learning for prior feature vectors of RSIT and texts.
3) Abundant experiments on the SYDNEY image–text dataset, UCM image–text dataset, RSICD image–text dataset, and RSITMD image–text dataset illustrate that the similarity between RSIs and texts is significantly boosted and the learned retrieval feature vectors contain better performance than other state-of-the-art cross-modal RSI–text (RSIT) retrieval algorithms.

The remaining sections of this article are organized as follows: Section II provides a comprehensive review of cross-modal retrieval in the context of RSI–sound (RSIS) and RSIT. In Section III, we delve into the intricate details of our proposed approach, MSITA. Section IV is dedicated to presenting the results of numerous ablation experiments and parameter studies conducted to evaluate the effectiveness of MSITA. Finally, in Section V, we offer concluding remarks summarizing the key findings and contributions of MSITA in the field of cross-modal retrieval. This restructuring aligns with a more logical and academically oriented flow.

II. Related Works

In this section, we will comprehensively review prior research related to cross-modal remote-sensing retrieval, focusing on two specific aspects: cross-modal RSIS retrieval and cross-modal RSIT retrieval.

A. Cross-Modal RSIS Retrieval

Cross-modal retrieval between RSIs and sounds involves searching for relevant sounds based on RSIs and vice versa. For instance, Mao et al. [35] pioneered the creation of three RSIS datasets and developed a deep network for cross-modal RSIS retrieval. Chen and Lu [36] devised a cross-modal RSIS hashing approach, which effectively selects triplets for hash code learning. Additionally, Guo et al. [37] introduced an image–sound alignment algorithm. Chen et al. [38] extended their work to capture multiscale contextual information from RSIs and sounds, enhancing retrieval performance. However, these methods often overlook semantic consistency. Addressing this limitation, Ning et al. [20] developed a semantics-consistent representation learning approach for cross-modal RSIS search. Chen et al. [39] proposed a quadruple-based hashing model to capture semantic information effectively for cross-modal RSIS retrieval.

B. Cross-Modal RSIT Retrieval

Cross-modal RSIT retrieval has garnered considerable attention due to its retrieval efficiency and versatile input options. Recent RSIT retrieval approaches predominantly rely on the semantic features of both images and texts, establishing a common cross-modal semantic space. This enables accurate matching between images and texts across different modalities. Abdullah et al. [40] introduced a novel text–image matching dataset and developed a bidirectional triplet model. Cheng et al. [31] established potential correspondence relationships between texts and their paired RSIs for cross-modal RSIT retrieval tasks. Yuan et al. [29] leveraged global and local features to construct a comprehensive multilevel RSIT matching model. Yuan et al. [41] proposed a lightweight multiscale retrieval algorithm, enabling the extraction of latent knowledge from different layers for cross-modal RSIT retrieval. While these cross-modal RSIT retrieval approaches have made notable progress, they still face challenges in capturing salient information, fine-grained correspondence relationships, and the underlying similarities between texts and RSIs. The quality of retrieval feature vectors is primarily determined by the learning of salient information and the similarity between these feature vectors, both of which are pivotal aspects of retrieval technology. In this article, we introduce a salient region learning module and a similarity learning objective function to enhance the similarity between retrieval feature vectors for texts and those for RSIs.

III. Proposed Method

In this section, we begin by presenting the problem definition for the MSITA approach, as outlined in Section III-A. Following that, we delve into the detailed process of learning features from RSIs in Section III-B. Section III-C elaborates on the text feature extraction process. In Section III-D, we provide a comprehensive explanation of the IGTA module. Finally, Section III-E introduces the objective function employed within the MSITA approach. The overall architecture of our proposed MSITA approach is depicted in Fig. 1. This restructuring aims to enhance the logical flow, academic style, and adherence to grammatical norms while ensuring clarity of expression.

A. Problem Definition

Given a set of N pairwise units denoted as $\mathcal{P} = \{I_m, T_m\}_{m=1}^{M}$, where I_m represents the mth RSI and T_m represents the corresponding text, our objective in the proposed MSITA approach is to learn feature vectors $F_{I'_m}$ for the RSIs and feature vectors $F_{T'_m}$ for the associated texts using a deep neural network. The primary goal of cross-modal retrieval is to discover an appropriate projection function for both RSIs and texts while preserving the similarity between images and their corresponding text descriptions. This facilitates effective retrieval of similar RSIs and texts when required.

B. Image Representation

Compared to natural images, RSIs exhibit variability in scale and orientation. RSIs typically contain a wealth of target information. For larger targets, global feature vectors are

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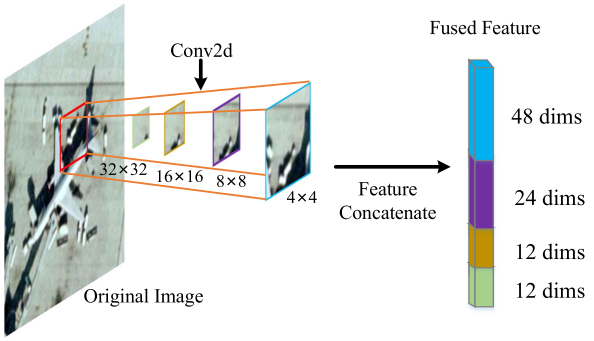


Fig. 2. Detailed process of the multiscale fusion module.

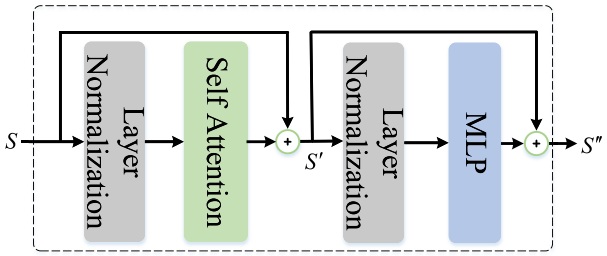


Fig. 3. Illustration of the attention structure.

more effective in representing them. However, when dealing with smaller targets in RSIs, the information related to these smaller targets tends to diminish or even disappear as the number of convolutional layers increases. Given the inherent complexity of RSIs, it becomes essential to consider multiscale fusion and saliency region learning techniques. These methods help in focusing on significant object information while simultaneously suppressing the expression of irrelevant or less useful information. The RSI branch is composed of several key components, including a multiscale fusion module, three transformer blocks, two patch merging stages, a salient learning module, and a fully connected layer. By incorporating these elements, we can enhance the capability of our model to handle the varying scales and rich target information present in RSIs, ultimately leading to improved performance and more effective feature representation.

First, to effectively capture finer details in RSIs, a multiscale fusion module has been designed to extract features from these images. The detailed process of the multiscale fusion module is illustrated in Fig. 2. This module takes an RSI as input, which is then subsampled using four kernels of different sizes (i.e., 4×4, 8×8, 16×16, and 32×32), all with the same stride. These kernels result in four corresponding blocks with the same center but different proportions. Subsequently, all four blocks are projected and combined to form the fused features.

Second, the transformer block comprises two attention structures. Each attention structure consists of a self-attention mechanism [42], two layers of layer normalization (LN), and a multilayer perceptron (MLP) layer, as depicted in Fig. 3. The detailed process can be outlined as follows: Beginning with a matrix S, it undergoes processing through the LN layer and a self-attention layer. Subsequently, the matrix S is added to the output of the self-attention layer, resulting in the matrix S'. Following this, S' undergoes further operations via the LN layer and the MLP layer. Finally, the matrix S' is added to the output of the MLP layer, producing the matrix S''.

Third, the patch-merging stages perform downsampling. This module serves to reduce the resolution, adjust channel numbers for hierarchical design, and conserve computational resources. It achieves this primarily by utilizing a convolution with a stride of 2 to decrease the resolution.

Fourth, to fully exploit salient information, the salient learning module is leveraged to select effective regions to form a new sequence. The salient learning module is designed in Fig. 4. We obtain image features I_{token} from the third transformer block. Image features I_{token} can be projected to Q, K, and V. Q, K, and V are input to a self-attention structure. And the similarity between Q and K can be written as

$$G = \text{Softmax}\left(\frac{QK^T}{\sqrt{d_{\text{model}}}}\right) \tag{1}$$

where $G = \{g_1, g_2, \ldots, g_U\}$ denotes the similarity between features of different regions, U represents the quantity of tokens, and d_{model} denotes embedding dimension. Then, we will use the fraction matrices G and V to compute the new image features. Finally, the new image features are fed into a layer paradigm, and a residual join is applied after the LayerNorm. The specific formula is written as

$$B = \text{Norm}(GV) + Q \tag{2}$$

where B is the new image feature. Norm denotes LN operation. Then, B is input to an MLP for linear transformation. Then, the features are midnormed with a LayerNorm. A residual connection is applied after the LayerNorm. The above process is expressed as

$$A = \text{Norm}(\text{MLP}(B)) + B \tag{3}$$

where A denotes feature embedding of information interaction attention. MLP denotes the MLP network.

Finally, the input data G and A are subjected to the Sequence Selection operation. This operation utilizes region relationship weights to determine the indices of the weighted regions, thereby selecting specific regions within the image, as illustrated in Fig. 4. The similarity scores G are sorted in descending order, resulting in the generation of the corresponding index set I. Subsequently, the top N tokens from A are chosen based on the index set I'' to extract the salient features V'. These salient features V' are then subjected to average pooling, yielding the ultimate image representation denoted as F_{I_m}.

C. Text Representation

When extracting text representations, we employ a bidirectional GRU and an MLP network to project the sentence into a d-dimensional space. This approach considers the contextual information of words during the encoding process rather than processing each word individually. For a given query text T_m consisting of J words, where T_m is represented as $\{o_j\}_1^J$,

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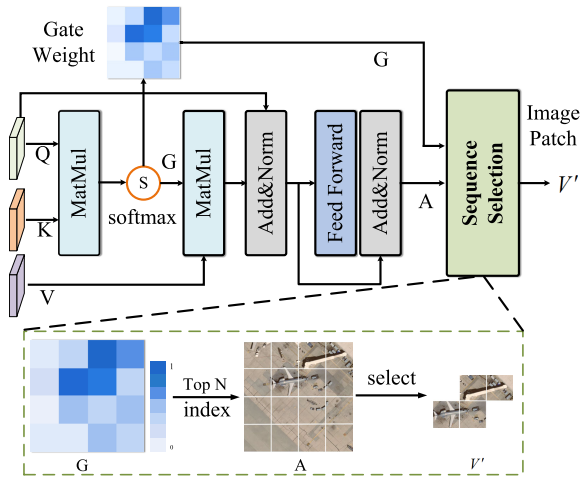


Fig. 4. Salient learning module.

we initially utilize a word embedding layer to map T_m using an embedding matrix. The process can be formulated as

$$x_j = W_G(o_j)(j \in [1, J]) \tag{4}$$

where x_j denotes word embedding vector. W_G denotes the embedding matrix. Then, the word embedding vector can be input into a bidirectional GRU network [43], [44], which can be summarized as follows:

$$\vec{h}_j = \vec{\text{GRU}}(x_j, \vec{h}_{j-1}), \quad j \in [1, J] \tag{5}$$

$$\overleftarrow{h}_j = \overleftarrow{\text{GRU}}\left(x_j, \overleftarrow{h}_{j-1}\right), \quad j \in [1, J] \tag{6}$$

where $\vec{h}_i$ represents hidden state from a forward GRU at step t. $\overleftarrow{h}_i$ represents a hidden state from a backward GRU at step t. Thus, the sentence feature can be formulated as

$$u_j = \frac{\vec{h}_j + \overleftarrow{h}_j}{2}, \quad j \in [1, J] \tag{7}$$

where u_j denotes the sentence feature, which is the average of the hidden forward and backward states. Finally, text representation F_{T_m} of text T_m can be obtained by projecting the sentence feature through a linear transformation.

D. Image-Guided Text Alignment

To utilize image information for guiding text alignment, we have introduced the IGTA module. This module is designed to incorporate image data to facilitate text alignment. An overview of the IGTA module is depicted in Fig. 1. The retrieval feature vector $F_{I'_m}$ for an RSI is obtained by projecting the image representation F_{I_m} through a linear transformation. The cosine similarity function is then employed to compute the similarity between these two modalities. The process can be formulated as

$$S(F_{I'_m}, F_{T_m}) = \frac{F_{I'_m} F_{T_m}}{\|F_{I'_m}\| \|F_{T_m}\|} \tag{8}$$

where F_{T_m} denotes text representation. $S(F_{I'_m}, F_{T_m})$ denotes the similarity between $F_{I'_m}$ and F_{T_m}. And then the sigmoid function is exploited to construct different attention weights according to the similarity between $F_{I'_m}$ and F_{T_m}

$$W^{\text{IT}} = \theta\left(S(F_{I'_m}, F_{T_m})\right) \tag{9}$$

where W^{IT} denotes the weight parameters. θ denotes the sigmoid function. The retrieval feature vector for texts can be obtained by projecting text representation through the weights W^{IT}

$$F_{T'_m} = F_{T_m} \times W^{\text{IT}} \tag{10}$$

where $F_{T'_m}$ denotes retrieval feature vector for texts. The IGTA can explore the fine-grained correspondence between image regions and texts.

E. Objective Function

To model the inherent similarity between RSIs and texts, the objective function of the MSITA approach comprises bidirectional triplet and InfoNCE terms. The architecture of the proposed MSITA approach is illustrated in Fig. 1. During the training phase, the bidirectional triplet term is employed to minimize the distance between matched pairs of RSIs and texts while maximizing the distance between mismatched pairs, as inspired by previous work [45]. Nevertheless, it is essential to acknowledge that there exists prior related information between image–text pairs. To capture this relatedness more effectively, we incorporate the InfoNCE term [46], leveraging the image representation F_{I_m} and text representation F_{T_m}, to learn the underlying similarity between RSIs and texts.

1) Bidirectional Triplet Term: Cross-modal learning involves the acquisition of appropriate projection functions for both RSI and text modalities while preserving the similarity between similar RSIs and texts. Instances sharing the same semantics should exhibit closer proximity in the projected space compared to instances with differing semantics. To achieve this objective, we introduce a bidirectional triplet term, ensuring that positive image–text pairs consistently yield higher similarity scores than negative image–text pairs. The bidirectional triplet term can be formulated as

$$\mathcal{L}_{\mathcal{B}} = \sum_{\hat{F}_{T'_m}} \left[\alpha - S(F_{I'_m}, F_{T'_m}) + S(F_{I'_m}, \hat{F}_{T'_m})\right]_+ + \sum_{\hat{F}_{I'_m}} \left[\alpha - S(F_{I'_m}, F_{T'_m}) + S(\hat{F}_{I'_m}, F_{T'_m})\right]_+ \tag{11}$$

where $\mathcal{L}_{\mathcal{B}}$ denotes the bidirectional triplet term. α denotes the margin parameter. $[x]_+ = \max(x, 0)$, and $S(F_{I'_m}, F_{T'_m})$ denotes the similarity between $F_{I'_m}$ and $F_{T'_m}$. The first sum can consider features $\hat{F}_{T'_m}$ of all negative texts for $F_{I'_m}$. The second sum can consider features $\hat{F}_{I'_m}$ of all negative images for $F_{T'_m}$.

2) InfoNCE Term: While the bidirectional triplet term effectively learns the similarity between the retrieval feature vectors $F_{I'_m}$ and $F_{T'_m}$, it does not capture the similarity between the image representation F_{I_m} and the text representation F_{T_m}. However, it is worth noting that there exists an inherent relationship between the features of the image–text pair. To capture the prior similarity between RSIs and texts by leveraging the image representation F_{I_m} and the

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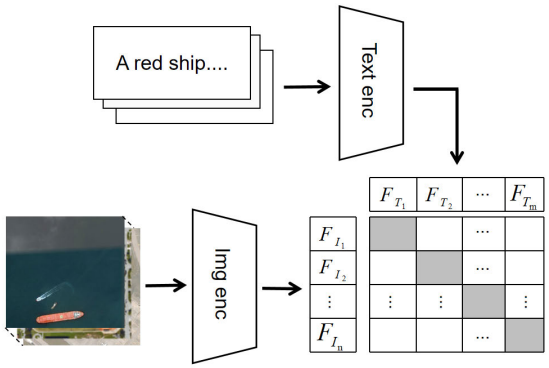


Fig. 5. Execution process of the InfoNCE term.

text representation F_{T_m}, we introduce the InfoNCE term, designed for contrastive learning. This term enhances the similarity between the image representation F_{I_m} and the text representation F_{T_m}. Furthermore, the InfoNCE term aims to maximize the lower bound of mutual information between a pair of variables. The execution process of the InfoNCE term is illustrated in Fig. 5. The InfoNCE term is given as

$$\mathcal{L}_{\mathcal{I}} = -\frac{1}{M}\sum_{n=1}^{M}\log\frac{\exp\left(S\left(F_{I_n}, F_{T_m}\right)/\tau\right)}{\sum_{m=1}^{M}\exp\left(S\left(F_{I_n}, F_{T_m}\right)/\tau\right)} \tag{12}$$

where $\mathcal{L}_{\mathcal{I}}$ denotes the InfoNCE term and τ denotes the learnable variable.

3) Total Objective Function: By considering the above two parts (i.e., the bidirectional triplet term $\mathcal{L}_{\mathcal{B}}$ and the InfoNCE term $\mathcal{L}_{\mathcal{I}}$), the total objective function is defined as

$$\mathcal{L} = \mathcal{L}_{\mathcal{B}} + \gamma\mathcal{L}_{\mathcal{I}} \tag{13}$$

where γ denotes the weighting parameter, $\mathcal{L}$ represents the total objective function, which can expand the distance between the instance and its corresponding negative instances, and reduce the distance between the instance and its positive instances as much as possible by considering the similarity between $F_{I'_m}$ and $F_{T'_m}$ and the similarity between F_{I_m} and F_{T_m}.

IV. Experiments

In this section, we will begin by describing the datasets, evaluation metrics, and experimental details. Subsequently, we will conduct an analysis and comparison between MSITA and state-of-the-art methods to demonstrate the effectiveness of our approach. Following that, we will perform a series of ablation experiments and parameter experiments. Finally, we will present the visualization results pertaining to retrieval effectiveness and the outcomes of IGTA.

A. Datasets and Evaluation Metrics

To evaluate the effectiveness of MSITA, our experiments utilize four commonly used RSIT datasets, which are the SYDNEY image–text dataset [47], the UCM image–text dateset [47], the RSICD image–text dateset [48], and the RSITMD image–text dateset [29]. Each image contains five different descriptive sentences in the four datasets. The RSICD image–text dataset is the largest number of sample RSIT datasets, which has 10 921 samples with size 224 × 224. The RSITMD image–text dataset has 4743 samples with size 256 × 256. The UCM image–text dataset has 2100 samples with size 256 × 256. The SYDNEY image–text dataset has 555 samples with size 500 × 500. For the UCM image–text dataset and the SYDNEY image–text dataset, we use 90% and 10% of the datasets as training sets and test sets. Following the setting of [29], we leverage 80%, 10%, and 10% of the datasets as a training set, validation set, and test set on the RSICD image–text dataset and the RSITMD image–text dataset, respectively.

We performed two types of image–text matching tasks. 1) Text retrieval is to retrieve true texts related to query image ($I \rightarrow T$) and 2) image retrieval is to retrieve true images related to query text ($T \rightarrow I$). The recall rate at K ($R@K$) is the commonly used evaluated metric of the retrieval task, which is defined as the percentage of queries containing real matches in the first k search results [49]. The higher the value of $R@K$ is, the better the performance is. Then, we utilized MR [50] to represent the average of multiple $R@k$, which intuitively denotes the overall performance of the approach.

B. Experiment Details

The proposed MSITA method is implemented using MindSpore and PyTorch. The experiments were conducted on a workstation equipped with a GeForce RTX 3090 GPU, an Intel Xeon[1] Silver 4210R CPU@2.40 GHz with 40 CPU cores, and 62.6 GiB of RAM. The loss function, as defined in (13), was optimized using the Adam optimizer with a learning rate of 2×10^{-4}, which was decayed by 0.7 every 20 epochs. The word vector's dimension was set to 300, and the batch size was fixed at 128. The embedding space for both images and texts was set to 512. We set the margin α to 0.2 and the parameter γ to 0.1. Additionally, the parameter τ was set to 0.1. The network was trained for 70 epochs, or training was stopped when the loss ceased to decline. To ensure the robustness of our experiments, we performed the same training and verification procedures five times for each experimental group and obtained average results.

C. Method Comparison

To evaluate the effectiveness of our MSITA method, we have conducted a comparative analysis by comparing it with several state-of-the-art approaches in the field of cross-modal retrieval. Below, we provide introductions to these methods.

1) *VSE++ [25]:* VSE++ extracts image and text features using VGG19 and bidirectional GRU, respectively. It employs a contrastive loss for model learning.
2) *SCAN [51]:* SCAN performs local feature extraction using a bottom-up attention mechanism, incorporating object detection. In this experiment, we compared both the text-to-image (t2i) and image-to-text (i2t) approaches developed in the SCAN framework.

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TABLE I
EXPERIMENTAL RESULTS BETWEEN MSITA AND COMPARED STATE-OF-THE-ARTS ON THE RSICD IMAGE–TEXT DATASET AND THE RSITMD IMAGE–TEXT DATASET

Method	RSICD dataset							RSITMD dataset						
	I→T			T→I			mR	I→T			T→I			mR
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
VSE++	3.38	9.51	17.46	2.82	11.32	18.1	10.43	10.38	27.65	39.6	7.79	24.87	38.67	24.83
SCAN t2i	4.39	10.9	17.64	3.91	16.2	26.49	13.25	10.18	28.53	38.49	10.1	28.98	43.53	26.64
SCAN i2t	5.85	12.89	19.84	3.71	16.4	26.73	14.23	11.06	25.88	39.38	9.82	29.38	42.12	26.82
CAMP-triplet	5.12	12.89	21.12	4.15	15.23	27.81	14.39	11.73	26.99	38.05	8.27	27.79	44.34	26.2
CAMP-bce	4.20	10.24	15.45	2.72	12.76	22.89	11.38	9.07	23.01	33.19	5.22	23.32	38.36	22.03
MTFN	5.02	12.52	19.74	4.90	17.17	29.49	14.81	10.40	27.65	36.28	9.96	31.37	45.84	26.92
LW-MCR	4.39	13.35	20.29	4.30	18.85	32.34	15.59	9.73	26.77	37.61	9.25	34.07	54.03	28.58
AMFMN	5.39	15.08	23.40	4.90	18.28	31.44	16.42	10.63	24.78	41.81	11.51	34.09	54.87	29.72
CLIP	7.96	21.41	32.48	5.42	20.02	33.19	20.08	14.89	33.55	45.64	9.81	32.68	47.43	30.67
MSITA	8.67	22.71	33.91	6.13	21.98	35.39	21.47	15.22	34.2	47.65	12.15	39.92	57.72	34.48

TABLE II
EXPERIMENTAL RESULTS BETWEEN MSITA AND COMPARED STATE-OF-THE-ARTS ON THE UCM IMAGE–TEXT DATASET AND THE SYDNEY IMAGE–TEXT DATASET

Method	UCM image-text dataset							SYDNEY image-text dataset						
	I→T			T→I			mR	I→T			T→I			mR
	R@1	R@5	R@10	R@1	R@5	R@10		R@1	R@5	R@10	R@1	R@5	R@10	
VSE++	12.38	44.76	65.71	10.1	31.8	56.85	36.93	24.14	53.45	67.24	6.21	33.56	51.03	39.27
SCAN t2i	14.29	45.71	67.62	12.76	50.38	77.24	44.67	18.97	51.72	74.14	17.59	56.9	76.21	49.26
SCAN i2t	12.85	47.14	69.52	12.48	46.86	71.71	43.43	20.69	55.17	67.24	15.52	57.59	76.21	48.74
CAMP-triplet	10.95	44.29	65.71	9.9	46.19	76.29	42.22	20.69	53.45	75.86	14.14	42.07	69.66	45.98
CAMP-bce	14.76	46.19	67.62	11.71	47.24	76.00	43.92	15.52	51.72	72.41	11.38	51.72	76.21	46.49
MTFN	10.47	47.62	64.29	14.19	52.38	78.95	44.65	20.69	51.72	68.97	13.79	55.51	77.59	48.05
LW-MCR	18.10	47.14	63.81	13.14	50.38	79.52	45.35	20.69	60.34	77.59	15.52	58.28	80.34	52.13
AMFMN	16.67	45.71	68.57	12.86	53.24	79.43	46.08	24.14	51.72	75.86	14.83	56.55	77.89	50.17
CLIP	13.81	47.14	62.38	9.9	46.19	73.14	42.1	31.03	53.45	74.13	21.03	55.52	73.44	51.44
MSITA	16.86	49.33	73.33	14.29	57.16	91.58	50.43	26.21	56.21	75.52	18.14	59.45	81.59	52.85

3) *CAMP [52]:* CAMP identifies key points in both images and texts, facilitating better interaction between them for improved image–text matching in cross-modal retrieval. In this experiment, we compared with both the CAMP-bce and CAMP-triplet approaches developed within the CAMP framework.
4) *MTFN [53]:* MTFN utilizes cross-modal embedding to perform tensor infusion, enhancing the modeling of relevance between different modalities. Additionally, it employs a reranking algorithm to improve model efficiency.
5) *LW-MCR [41]:* LW-MCR introduces a lightweight multiscale retrieval algorithm to acquire dark knowledge from different layers.
6) *AMFMN [29]:* AMFMN employs a multiscale and redundant information filtering structure for processing high-level image features. The information from both images and texts is aligned through a semantic alignment structure.
7) *CLIP [41]:* CLIP is a cross-modal pretrained model designed for contrast-based picture–text learning. We use CLIP in conjunction with a pretrained ResNet50 model for RSIT retrieval tasks.

These comparisons are conducted to assess the performance of our MSITA method in the context of existing state-of-the-art approaches.

1) Results on the RSICD Image–Text Dataset: The cross-modal RSIT retrieval results, comparing our proposed MSITA method with other approaches, are presented in the left portion of Table I. Notably, while other cross-modal retrieval algorithms exhibit strong performance, the MSITA algorithm we propose achieves the highest average retrieval accuracy. Compared to state-of-the-art methods, our approach demonstrates an improved accuracy of 10.51 for R@10 in text retrieval and 3.95 in image retrieval. These comparative results underscore the significant enhancement in cross-modal retrieval achieved by the MSITA method. This notable improvement can be attributed to several key factors. First, the MSITA algorithm excels in capturing salient information within the data. Second, it leverages IGTA, facilitating a more robust cross-modal alignment process. Last, it effectively learns the prior similarity between RSIs and texts, further contributing to the success of cross-modal RSIT retrieval. These findings underscore the efficacy of the MSITA method and its ability to enhance cross-modal retrieval performance substantially.

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2) Results on the RSITMD Image–Text Dataset: Cross-modal RSIT retrieval results between our proposed MSITA method and other state-of-the-arts by using RSIs (i.e., texts) to retrieve texts (i.e., RSIs) on the RSITMD image–text dataset is displayed in the right of Table I. Obviously, it can be seen from Table I that the MSITA approach achieves the best performance than VSE++, SCAN t2i, SCAN i2t, CAMP-triplet, CAMP-bce, MTFN, LW-MCR, and AMFMN. For text retrieval, the MSITA approach enhances R@1 to 15.22% from 10.38% conducted by VSE++, 10.18% conducted by SCAN t2i, 11.06% conducted by SCAN i2t, 11.73% conducted by CAMP-triplet, 9.07% conducted by CAMP-bce, 10.40% conducted by MTFN, 9.73% conducted by LW-MCR, and 10.63% conducted by AMFMN. For image retrieval, the MSITA approach enhances R@1 to 12.15% from 10.1% conducted by VSE++, 9.82% conducted by SCAN t2i, 8.27% conducted by SCAN i2t, 5.22% conducted by CAMP-triplet, 9.96% conducted by CAMP-bce, 9.25% conducted by MTFN, 11.51% conducted by LW-MCR, and 10.63% conducted by AMFMN. The retrieval results clearly demonstrate the effectiveness of the proposed cross-modal RSIT retrieval algorithm. Because the MSITA approach can learn salient information and fine-grained correspondence relationships to tackle the issue of RSI and text alignment. Moreover, the MSITA approach can perverse the similarity of different modalis and reinforce the prior similarity between RSIs and texts.

3) Results on the UCM Image–Text Dataset: The table presented in the left section of Table II displays the cross-modal RSIT retrieval results for our proposed MSITA method in comparison to other methods. While LW-MCR achieves the highest R@1 metric in sentence retrieval, it is noteworthy that the MSITA method outperforms other methods across most of the metrics.

4) Results on the SYDNEY Image–Text Dataset: The cross-modal RSIT retrieval results for the UCM image–text dataset are presented on the right side of Table II. Similar observations can be made for the UCM image–text dataset. While LW-MCR achieves the best metrics in sentence retrieval for R@5 and R@10, it is noteworthy that the proposed MSITA method outperforms LW-MCR across most metrics, demonstrating the effectiveness of the MSITA method. This highlights the superior performance of the MSITA method in the context of cross-modal RSIT retrieval.

D. Ablation Experiments

To evaluate the effectiveness of MSITA, we conduct the following experiments: First, we remove the IGTA module and the InfoNCE term for model training (i.e., MSITA-IGTAInfoNCE). Second, we leverage ResNet18 to replace the multiscale fusion module and the transformer with a salient region learning module (i.e., MS) for the model training (i.e., MSITA-MS). Third, we perform image feature extraction with ResNet18 and remove the InfoNCE term (i.e., MSITA-MSInfoNCE). Fourth, we implement cross-modal remote-sensing retrieval without exploiting IGTA (i.e., MSITA-IGTA). Fifth, we realize the proposed MSITA method without the InfoNCE term learning (i.e., MSITA-InfoNCE). Finally, we realized the proposed MSITA method. Table III displays the comparison of sentence retrieval and image retrieval on the RSITMD dataset for MSITA-IGTAInfoNCE, MSITA-MS, MSITA-MSInfoNCE, MSITA-IGTA, MSITA-InfoNCE, and MSITA. It can be noticed from Table III that the proposed MSITA method can achieve better performance than MSITA-IGTAInfoNCE, MSITA-MS, MSITA-MSInfoNCE, MSITA-IGTA, and MSITA-InfoNCE. For example, SITA can improve MR from 30.64% conducted by MSITA-IGTAInfoNCE, 31.7% conducted by MSITA-MS, 30.73% conducted by MSITA-MSInfoNCE, 30.17% conducted by MSITA-IGTA, and 34.14% conducted by MSITA-InfoNCE to 34.48%. This is because the proposed MSITA method utilizes IGTA, the InfoNCE term learning, the multiscale fusion module, and the transformer with salient region learning module to implement cross-modal remote-sensing retrieval effectively.

To validate the effectiveness of image feature extraction network, we compare the proposed image feature extraction network with some commonly used image feature extraction networks (i.e., ResNet18 [54], ResNet50 [54], ResNet101 [54], ViT [55], and Swim-T [56]). Cross-modal RSIT retrieval results with different network architectures on the RSITMD image–text dataset are shown in Table IV. The experimental results show that the designed network architecture can learn better RSI features than ResNet18, ResNet50, ResNet101, ViT, and Swim-T.

To evaluate the effectiveness of the alignment way, we utilize some existing alignment ways to combine with our model for the experimental comparison. Existing alignment ways contain *fusion attention* (FA) [29], *semantic alignment module* (SAM) [31], *similarity attention* (SIA) [29], and *soft attention* (SOA) [57]. Cross-modal RSIT retrieval results with different alignment ways on RSITMD image–text dataset is shown in Table V. Although other alignment ways can achieve certain effects, we can find that the improved accuracy of the proposed alignment way is obvious. It can be concluded that the proposed alignment way is effective.

To verify the robustness of the proposed method, we add Gaussian noise to the RSIs of four datasets. First, we add Gaussian noise to the RSITMD image–text dataset for the model training (i.e., RSITMD + G). Second, we add Gaussian noise to the UCM image–text dataset for experimentation (i.e., UCM + G). Third, we add Gaussian noise to the RSICD image–text dataset for the model training (i.e., RSICD + G). Fourth, we add Gaussian noise to the SYDNEY image–text dataset for experimentation (i.e., SYDNEY + G). Cross-modal RSIT retrieval results on four image–text datasets are shown in Table VI. The experimental results show that the accuracy of the proposed method has little change after adding the standard Gaussian noise, which can demonstrate the robustness of the proposed method.

To assess the performance of various loss functions, we conducted a comparative analysis and employed a range of existing loss functions. In Table VII, we have outlined several different loss functions, including the NCE loss [58], InfoNCE loss [46], and an enhanced hyperbolic NCE (HNCE) loss [59].

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TABLE III
CROSS-MODAL RSIT RETRIEVAL RESULTS WITH DIFFERENT FACTORS ON THE RSITMD DATASET

Ablation	MS	IGTA	InfoNCE	I→T			T→I			mR
				R@1	R@5	R@10	R@1	R@5	R@10	
baseline	×	×	×	10.84	29.42	42.69	10.61	33.49	38.66	29.60
MSITA-IGTAInfoNCE	✓	×	×	13.76	30.30	43.14	11.37	35.08	50.26	30.64
MSITA-MS	×	✓	✓	14.12	30.7	43.05	11.27	37.12	53.95	31.7
MSITA-MSInfoNCE	×	✓	×	12.78	30.62	42.57	11.03	35.73	51.65	30.73
MSITA-IGTA	✓	×	✓	9.86	26.9	40.39	9.38	33.91	53.66	29.02
MSITA-InfoNCE	✓	✓	×	15.26	33.05	46.41	12.33	39.19	57.54	33.96
MSITA	✓	✓	✓	15.22	34.2	47.65	12.15	39.92	57.72	34.48

TABLE IV
CROSS-MODAL RSIT RETRIEVAL RESULTS WITH DIFFERENT NETWORK ARCHITECTURES ON THE RSITMD IMAGE–TEXT DATASET

Backbone	I→T			T→I			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
ResNet18	14.12	30.7	43.05	11.27	37.12	53.95	31.7
ResNet50	13.31	32.47	45.13	11.47	37.89	53.61	32.31
ResNet101	14.11	32.92	46.06	12.53	37.38	53.92	32.82
ViT	13.4	31.46	44.95	11.47	36.91	53.61	31.97
Swin-T	13.42	31.78	45.21	11.49	37.45	54.51	32.31
MSITA	15.22	34.2	47.65	12.15	39.92	57.72	34.48

TABLE V
CROSS-MODAL RSIT RETRIEVAL RESULTS WITH DIFFERENT ALIGNMENT WAYS ON THE RSITMD IMAGE–TEXT DATASET

Alignment	I→T			T→I			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
FA	12.52	29.33	41.76	9.76	35.54	54.3	30.54
SAM	11.24	30.35	43.94	10.93	35.13	53	30.77
SIA	13.27	30.75	44.02	11.37	37.07	55.66	32.02
SOA	12.69	32.38	45.13	11.85	38.98	56.71	32.96
MSITA	15.22	34.2	47.65	12.15	39.92	57.72	34.48

TABLE VI
CROSS-MODAL RSIT RETRIEVAL RESULTS ON FOUR IMAGE–TEXT DATASETS

Data	I→T			T→I			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
RSITMD	15.22	34.2	47.65	12.15	39.92	57.72	34.48
RSITMD+G	14.15	32.52	44.13	12.19	37.03	52.10	32.02
UCM	16.86	49.33	73.33	14.29	57.16	91.58	50.43
UCM +G	13.33	47.14	70.79	12.38	53.74	91.65	48.17
RSICD	8.67	22.71	33.91	6.13	21.98	35.39	21.47
RSICD+G	7.56	21.04	31.29	5.44	20.70	33.90	19.99
SYDNEY	26.21	56.21	75.52	18.14	59.45	81.59	52.85
SYDNEY+G	20.11	52.29	70.68	21.71	60.45	79.42	50.78

Based on the experimental results, it can be concluded that the combination of loss functions we proposed demonstrates significant effectiveness.

E. Parameter Analysis

Tables VIII and IX reveal the impact of different parameters α and γ on cross-modal RSIT retrieval performance. For

TABLE VII
CROSS-MODAL RSIT RETRIEVAL RESULTS WITH DIFFERENT LOSS FUNCTIONS ON THE RSITMD IMAGE–TEXT DATASET

Loss	I→T			T→I			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
NCE	15.26	35.39	44.46	12.47	37.56	55.39	33.42
HNCE	14.60	32.85	45.13	12.32	40.11	57.23	33.70
InfoNCE	15.22	34.2	47.65	12.15	39.92	57.72	34.48

TABLE VIII
CROSS-MODAL RSIT RETRIEVAL RESULTS WITH DIFFERENT MARGIN PARAMETER α ON THE RSITMD IMAGE–TEXT DATASET

α	I→T			T→I			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
0.1	14.74	33.7	46.53	11.57	38.2	56.34	33.51
0.2	15.22	34.2	47.65	12.15	39.92	57.72	34.48
0.3	14.82	33.71	46.85	12.1	39.65	57.28	34.07
0.4	12.5	31.96	43.58	10.7	38.62	59.66	32.84
0.5	9.07	24.92	39.15	7.47	32.87	56.4	28.31
0.6	8.84	24.33	38.27	6.76	32.43	55.35	27.66
0.7	6.34	19.83	35.32	5.17	26.53	49.77	23.83
0.8	7.34	22.69	35.13	6.69	29.89	54.8	26.09
0.9	4.42	13.35	23.45	3.19	15.40	28.55	14.72
1.0	2.87	9.51	17.25	2.47	9.95	20.30	10.39

Table VIII, as α varies between 0.1 and 0.3, the retrieval performance in both i2t ($I \rightarrow T$) and t2i ($T \rightarrow I$) tasks gradually improves, indicating that smaller α values contribute to better retrieval results. The overall mean retrieval rate (mR) also increases with decreasing α, suggesting that smaller α values generally lead to improved cross-modal retrieval performance. Regarding Table IX, in the i2t ($I \rightarrow T$) retrieval task, R@1, R@5, and R@10 gradually increase with the rise of γ until reaching a peak, after which there is a slight decline. This suggests that moderate γ values perform best in this task. Similar trends are observed in the t2i ($T \rightarrow I$) retrieval task, where moderate γ values yield the best results. The overall mean retrieval rate (mR) reaches its peak at smaller γ values and then experiences a slight decrease. In summary, these experimental results indicate that model performance is optimal within a moderate range of α and γ parameters. However, it is essential to adjust these parameters based on

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Fig. 6. Top-5 retrieval texts with the given some query images on the four datasets. The right retrieval texts are annotated with green. The wrong retrieval texts are annotated with red.

TABLE IX
CROSS-MODAL RSIT RETRIEVAL RESULTS WITH DIFFERENT PARAMETER γ ON THE RSITMD IMAGE–TEXT DATASET

γ	I→T			T→I			mR
	R@1	R@5	R@10	R@1	R@5	R@10	
0	15.26	33.05	46.41	12.33	39.19	57.54	33.96
0.1	15.22	34.2	47.65	12.15	39.92	57.72	34.48
0.3	16.68	33.67	45.97	12.56	39.81	57	34.28
0.5	14.38	33.33	47.41	13.06	39.55	55.92	33.94
0.7	15.53	34.55	46.94	12.42	39.22	56.17	34.14
0.9	15.7	34.24	45.3	12.41	38.79	56.13	33.76

specific datasets and tasks, as their performance impact may vary under different circumstances.

In Table X, we have compared the model parameter counts for various methods. The proposed MSITA approach has 48.14 million parameters, which is about one-third of the CLIP method. Compared with other methods, the proposed model achieves good results while requiring a relatively lower number of parameters.

F. Visualized Results

Fig. 6 presents the qualitative outcomes of text retrieval when provided with an RSI query across the four datasets. For each query image, we have showcased the top-5 retrieval results. The correct retrieval texts are indicated in green, while the incorrect ones are highlighted in red. Fig. 7 illustrates the qualitative results of image retrieval when given a query text across the four datasets. We display the top-5 retrieval images, ranking them from left to right. Correct retrieval

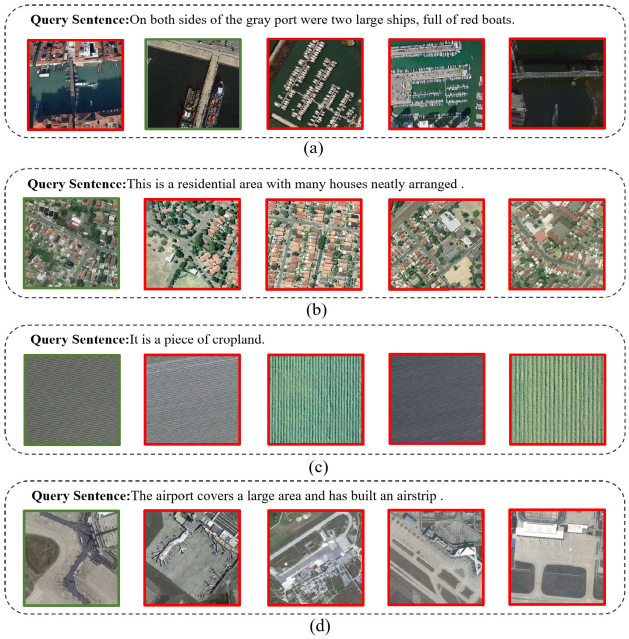


Fig. 7. Top-5 retrieval images with the given query texts on the four datasets. The right retrieval images are annotated with green. The wrong retrieval images are annotated with red. (a)–(d) Top-5 retrieval images for the given query texts.

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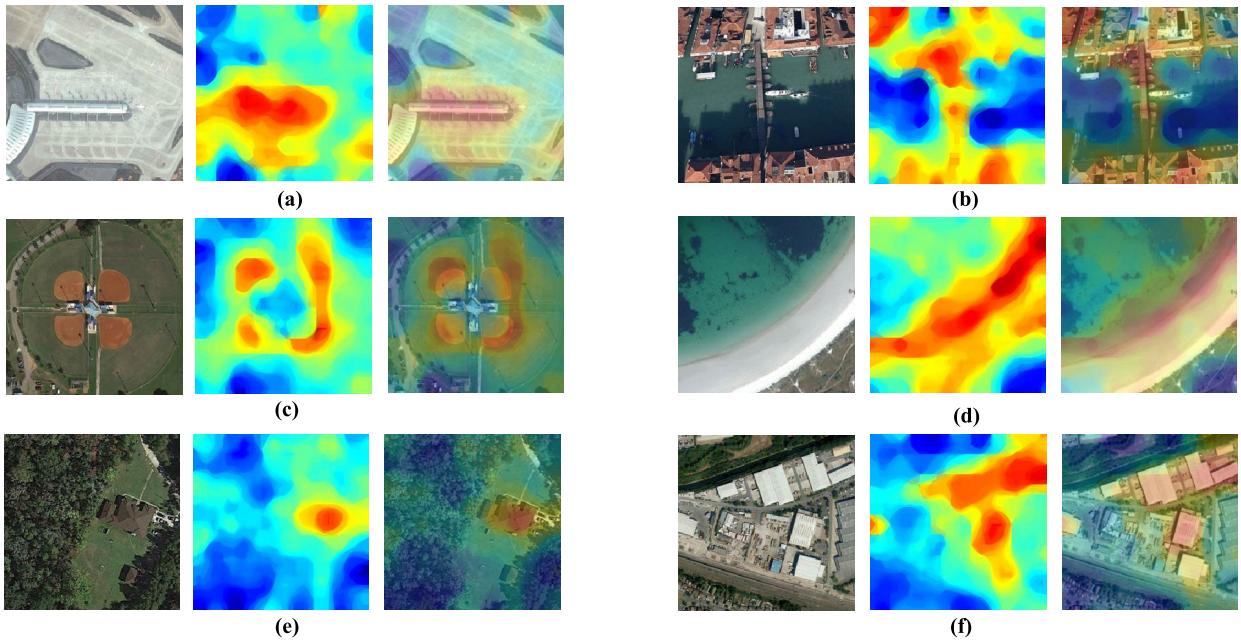


Fig. 8. Salient location visualization. (a)–(f) Visible results of six typical RSIs. (Left) Original images. (Middle) Salient mask. (Right) Superimposed images.

TABLE X
COMPARISON OF PARAMETERS OF VARIOUS APPROACHES

Approach	VSE++	MTFN	LW-MCR	AMFMN	CLIP	MSITA
Pramaters(M)	15.78	77.90	1.65	35.94	149.62	48.14

images are denoted in green, while incorrect ones are marked in red. Observing Figs. 6 and 7, it is evident that the proposed MSITA method effectively retrieves most of the correct results, often ranking them at the top, even in complex scenarios. However, some inaccuracies are noticeable in both Figs. 6 and 7. In Fig. 6, when retrieving texts related to ships, it is apparent that our results contain descriptions exclusively about ships. Nevertheless, a discrepancy in color resulted in a text retrieval error in the fifth rank. In Fig. 7(a), the first incorrect image bears a resemblance to the second correct image. The prominently red portion in the first image is quite conspicuous. The MSITA method may have focused more on this region, leading to retrieval errors. In summary, our visualized results indicate that the proposed MSITA method excels in identifying the relevance between images and texts, but occasional errors may still occur.

G. Salient Location Visualization

Fig. 8 illustrates the salient location visualization for six representative RSIs. For instance, in Fig. 8(b), the model's attention is directed toward bridges, boats, and red houses. However, it is worth noting that the attention on the red house appears to be somewhat limited in scope. This suggests that the proposed model tends to prioritize focusing on the closely related parts between objects during salient analysis. In Fig. 8(d), the model exhibits a stronger focus on the beach areas that are in close proximity to the ocean. Similar to the observation in Fig. 8(b), it can be inferred that the model emphasizes the closely associated regions. Moving to Fig. 8(f), the model effectively identifies the white factory but not the gray one. This may be attributed to the model potentially classifying the gray factory as a road. Collectively, these experiments provide evidence that the proposed MSITA method demonstrates a capability to emphasize the salient regions within RSIs, although further improvements in certain cases may be warranted.

V. CONCLUSION

In this article, we present a novel method for cross-modal RSIT retrieval. This method is designed to capture salient information and establish a prior similarity between RSIs and textual descriptions. First, our proposed MSITA method incorporates a multiscale fusion module and a salient region learning module to effectively identify salient regions within RSI. Second, we introduce a straightforward, yet efficient alignment approach. This approach utilizes temporal information to guide the alignment process between image features and text features. Third, we define an objective function aimed at computing similarity scores for retrieval feature vectors and facilitating similarity learning for the prior feature vectors associated with RSIs and texts. Ultimately, we validate the rationale and effectiveness of the MSITA method through comprehensive experiments on various benchmark datasets. Looking ahead, our future work will revolve around two primary aspects. First, we plan to explore the applicability of MSITA in other domains, such as cross-modal biometric feature matching, to highlight its versatility and effectiveness. Second, we intend to delve into the one-to-many phenomenon in RSIs and texts, aiming to enhance the performance of cross-modal retrieval in the realm of remote sensing.

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